

The Containment Hypothesis in DM (Bobaljik, 2012)

A comprehensive account

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Distributed Morphology (**DM**) is a theory which was introduced by the linguists Morris Halle and Alec Marantz (Halle and Marantz, 1993). It was further refined by their students Noyer (Noyer, 1997), Embick (Embick, 1997) and Bobaljik (Bobaljik, 2012). In DM, as the derivation exits the syntax, the terminal nodes (ABSTRACT MORPHEMES) contain only featural content; there is no phonological material at this phase. Only when the Vocabulary Insertion competition is resolved by means of Maximal Subset Condition can a terminal node be mapped to a phonological form (Exponence). In case several Vocabulary items meet the conditions for insertion, the item matching the greatest number of features specified in the terminal morpheme is selected ("it wins"). The lexicon in DM is tri-modular consisting of three lists: The formative list (ROOTs), the Vocabulary List (VI) and the Encyclopedia List (Idiosyncratic meaning storehouse). DM specifies that **Spellout** takes place when terminal nodes are *mapped* onto their PF and checked for LF interpretation when it comes to Idiomatic decomposition. In fact, there are three core properties that mark DM: **Late Insertion**, **Underspecification**, and **Syntactic Hierarchical Structure All the Way Down**. Central to Suppletive allomorphy is the **Containment Hypothesis**, a DM-based theory introduced and elaborated by Bobaljik (Bobaljik, 2012). It offers promising insights on the *regular and suppletive allomorphy* of comparatives, superlatives and Change-of-state verbs in DM.

The containment hypothesis hinges on the Comparative Superlative Generalization (CSG). CSG states that the representation of the superlative properly contains that of the comparative (in all languages that have a morphological superlative). (Bobaljik, 2012, p.49). There are five possible morphological Exponence patterns here, two of which are unattested: **AAA**, **ABB**, **ABC**, ***ABA** and ***AAB**. While **A** refers to the default form (positive) of the adjective, **B** refers to the comparative form. **C** refers to a *suppletively independent* form of the superlative. The latter is *very rare* across languages. So, we will focus on generalization **ABB**: when the comparative has a *suppletive* form, the superlative *will also* be *suppletive*, and vice versa. This generalization seems promising as it allows the morphological patterns **AAA**, **ABB** and **ABC** and rules out the ***ABA** paradigm (**kabi:r – ?akbar – ?akabi:r*) and ***AAB** (**kabi:r – kabi:r – ?akbar*). Figure 1 illustrates this with examples from English.

	POS	CMR	SPRL	
a. Regular	A	A	A	big – bigger – biggest
b. Suppletive	A	B	B	good – better – best
c. Doubly suppletive	A	B	C	bonus – melior – optimus
d. Unattested	A	B	A	*good – better – goodest
e. Unattested	A	A	B	*good – gooder – best

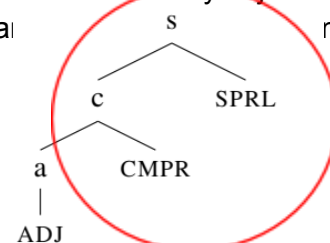
The derivation of superlatives from comparatives is also evident in languages where superlatives are formed by means of a **definiteness marker**. One superlative-forming strategy involves the addition of a **definite article** (or other **definiteness marker**) to the comparative (whether the comparative is itself formed morphologically or periphrastically) (Bobaljik, 2012, p.53). In fact, a comparative construction involves no overt marking on the adjective as in **French** and **Standard Arabic**. The examples below typify the way superlatives are derived in these two languages (See figure 2)

Data from Modern Standard Arabic			
Positive	comparative	superlative	
* ðakij	?aðka:	?al-?aðka:	
* yaniy	?ayna	?al-?ayna	
* kabir	?akbar	?al-akbar	
* Sari:ʕ	?asraʕ	?al-?asraʕ	

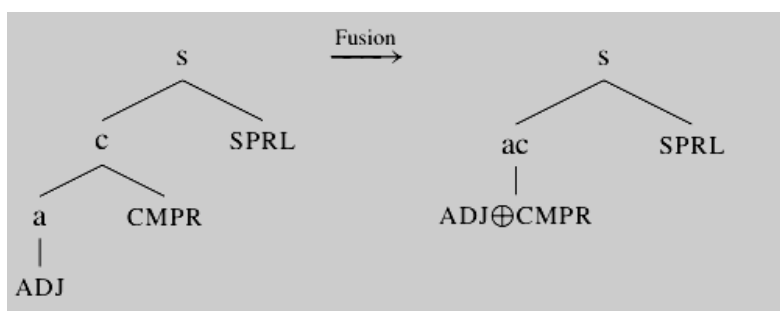
Data from French			
Positive	comparative	superlative	
Mauvais - plus mauvais	-	le plus mauvais	(bad)
gros - plus gros	-	le plus gros	(big)

As shown above, Arabic and French have **no** morphological distinction between **comparative** and **superlative**, and the only superlative-marker is the **elative** form used in superlative contexts (French: **le/lalles** and Arabic: **ʔal-**) (Bobaljik, p. 53). However, this does not undermine the containment hypothesis, especially the CSG; indeed, the data above supports the ABB pattern quite consistently. All superlative forms in French and Arabic **contain** the comparative form. Note that the containment relation holds in terms of **syntactic structural adjacency**: the comparative head is dominated by the superlative and so it is merged with it. In terms of Exponence, the resultant form must contain the comparative form, because the superlative is structurally adjacent enough to the node COMPR (Seddiqi, 2012). This is contrasted in a) and b) in figure 3.

- | | |
|----|--|
| a) | $\sqrt{\text{good}} \rightarrow \text{better} / \text{-----}] \text{Cmpr}]$ |
| b) | $\sqrt{\text{good}} \rightarrow * \text{goodest} / \text{-----}] \text{SPRL}]$ |



The **adjacency condition** on suppletion shown in figure 3 is motivated within the current theory by the work it does in explaining the **nonoccurrence** of AAB patterns (**good – gooder – best*) (Bobaljik, 2012, p.144). In the case of **AAA**, that comparatives contain the default form of the adjective (Positive) is a result of Fusion of the two **terminal nodes** (a) and (c), so that the fused node is mapped to a PF that **necessarily** contains both the positive form and the comparative form as represented in figure 4 below:



However, the issue of synthetic forms for the comparative and superlative (-er, est- st, ier....) and free morphemes (more, most) has been densely debated by Embick, Noyer and Bobaljik (Embick, 2007, Bobaljik, 2012). Bobaljik considers that **Local Dislocation** cannot deal with suppletion. The synthetic comparatives and superlatives of adjectives such *good, bad, little and many/much* are often **suppletive**. Local Dislocation rule (Embick 2007a) contains a reference to the **phonological form** of the adjective in question; so, the adjectival stem must be **spelled out** before combining with the comparative/superlative suffix, which incorrectly predicts the stem suppletion in the case of *good better best*, hence the exclusion of **ABA* and **AAB*. Bobaljik presumes that the containment hypothesis has important implications for language acquisition as well. How a learner comes to know the comparative and superlative form for a given root (e.g. *polite – politer ~ more polite*) depends on convergent sources of evidence available to the learner; these **recurrent forms** refer to **Positive evidence** in the **input**. Yet, the derivation of synthetic comparatives and superlatives by any form of lowering, including **Affix Hopping**, is refuted by examples from **coordinated comparatives** as in (5) (Jackendoff, 2000):

- (5)
- a) *more and more beautiful*
 - b) *prettier and prettier*

Indeed, Bobaljik's analysis of containment relations is interesting because it has also covered the morphology of the so-called **Δ-verbs** (**deadjectival verbs**) across a substantial number of languages. Further evidence in support of the ABB pattern and against the **ABA* paradigm can be found in the derivation of Change-of-state Verbs (Δ-verbs) whose derivatives contain the comparative form of the adjective. Bobaljik introduced another generalization in the containment hypothesis. He argues for the existence of **an abstract**

morpheme V Δ which may **only combine with comparative stems** (Bobaljik, 2012, p. 187). The charts¹ below illustrate the C Δ G concretely.

Table 6.1
Deadjetival verbs from suppletive adjectives

C Δ G	LOSS	ADJ	CMPR	VERB	LANGUAGE; COGNATES
	good	good	bett-er	bett-er	English; other Germanic
	good	dobry	lepsz-y	po- lepsz-yć	Polish; Czech
	good	xoroš-ij	luč-š-e	u- luč-š-at'	Russian

Therefore, the containment hypothesis may be thought of as a “**Filter to Exponence**” when it comes to VI pertaining to comparatives, superlatives (CSC) and Δ -verbs (See C Δ G CHART above). Consider the category-changing *contextual* allomorphy **less[A]→lessen/-----[Δ V]**. The Δ -verb **lessen** contains the *comparative* form **less** rather than the **positive form** “few”. Therefore, the form ***fewen** cannot surface as it violates the C Δ G condition. In this regard, Bobaljik notes that verbs which do not denote a **change of state** (such as coming to have a property), also called “**ascriptive**” verbs, with meanings such as **belittle** and **exaggerate** (ibid).

Interestingly, In **American Sign Language** as well, the morphological superlative appears to contain the morphological comparative (although only a small set of adjectives have morphological grades): the sign glossed -EST (a suffix) consists of the -ER sign but uses a greater extent of movement, suggesting the superlative could be analyzed as the comparative plus an intensifier (Bobaljik, 2012, p. 50). Bobaljik extends the analysis to include the regular allomorphy of tense inflectional forms: **Present-Participle-Preterite**. He argues that they are relevant for the containment analysis. The features characterizing (the suffix node of) the participle are **contained** in those for the more *marked preterite*. He notes that “the [PAST] and [FINITE] features are both **adjacent to the root** (ibid, p. 161). Examples of AAA pattern are abundant in English and many other languages (e.g. **Sing-sang-sung; ride, rode, ridden**).

This all comes down to the fact the containment hypothesis remains a powerful toolbox in DM, as it accounts for a wide range of suppletive allomorphy cross-linguistically, especially that of comparatives, superlatives and certain verbs, cross-linguistically. Bobaljik’s two containment generalizations CSG and C Δ G remain strongly consistent in a large corpus of attested data in more than 300 languages of the world, and this is evidence of the resiliency of the containment hypothesis. The morphology of superlatives in Arabic and French operates by adjoining the **definiteness node** which c-commands the superlative node, itself a fused form of two nodes (CMPR and SPRL) where the only Exponence needed is a definiteness article. Finally, we have also seen how Δ -verbs contain the comparative form, rather than the **positive form**(few-less-**lessen**/*fewen). Bobaljik’s containment hypothesis has also been researched in terms of its pertinence to Syntactic Hierarchy underlying the morphology of words. We have seen that Superlatives derive from Positive forms due to the Structural Adjacency Condition. However, this does not mean the hypothesis is flawless. Indeed, there are prosodic constraints that do interfere with containment relations posited by Bobaljik (Halle, 2013, p. 62)

¹ Bobaljik, 2012, p. 185

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